Supplementary Information

Disentangling Between-Target and Within-Target Signal in PPI $\Delta\Delta G$ Benchmarking

Supplementary Note 1 | Derivation of the covariance decomposition

Write the true value and prediction for mutation i in target c as $y_{ci} = \mu_c + \varepsilon_{ci}$ and $\hat{y}_{ci} = \hat{\mu}_c + \hat{\varepsilon}_{ci}$. Expanding $\mathrm{Cov}(y, \hat{y})$ gives a between-target term and a within-target term,

$$\mathrm{Cov}(y, \hat{y}) = \frac{1}{N} \sum_c n_c \, (\mu_c - \bar{\mu}) \left(\hat{\mu}_c - \bar{\hat{\mu}}\right) + \frac{1}{N} \sum_c \sum_i \varepsilon_{ci} \, \hat{\varepsilon}_{ci}; \tag{1}$$

all cross terms vanish identically because within-target deviations have zero mean in every target. Dividing by $\sigma_y \sigma_{\hat{y}}$ gives C_{B} and C_{W} and establishes

$$r_{\text{pooled}} = C_{\mathrm{B}} + C_{\mathrm{W}} \tag{2}$$

exactly. The identity residual is reported as a quality-control quantity (observed maximum 3.331×10^{-16}). Because C_{B} and C_{W} share the same denominator they are covariance-signal contributions on a common scale, not independent correlations.

Supplementary Note 2 | Row universes and their relationship

The controlled-split universe (7,523 rows, 230 complexes, 112 structural families) is the full set of forward-paired true $\Delta\Delta G$ observations. It contains 4,812 distinct (complex, mutation) keys; 2,011 keys carry more than one experimental record and 1,405 of those disagree in $\Delta\Delta G$.

The annotation namespace (4,473 rows, 214 targets, 108 structural families) is the feature-complete subset of that universe: every one of its 4,473 rows is present in the controlled-split universe, and 3,050 rows of the latter are absent from it (16 targets and 4 structural families are not represented). The two row counts therefore differ because they describe different filtering stages of the same underlying data, not different benchmarks.

The target-count distributions are:

Universe	$n \geq 1$	$n \geq 2$	$n \geq 3$	$n \geq 5$	$n \geq 10$
Controlled-split (7,523)	230	203	193	172	118
Annotation (4,473)	214	180	164	130	85

Supplementary Note 3 | Matched universes and row-set fingerprints

Each prediction output received a SHA-256 fingerprint of its sorted row identifiers. Cross-model comparison was performed only between outputs sharing a fingerprint; raw metrics were never compared across fingerprints.

Table S1: Matched-evaluation strata and row-set fingerprints.

Universe	Artifact	Row-set fingerprint	Rows (N)	Targets	Families	Targets ($n \geq 3$)	Targets ($n \geq 5$)	Targets ($n \geq 10$)	Models
U_V6_2_split4_8d003e00	V6_2_split4	8d003e00057d932a	7523	230	112	193	172	118	36
U_V6_2_1_donly_8d003e00	V6_2_1_donly	8d003e00057d932a	7523	230	112	193	172	118	6
U_V7.1_z_34b45937	V7.1_z	34b45937b42b7114	4473	214	108	164	130	85	10
U_V7.0_ablation_34b45937	V7.0_ablation	34b45937b42b7114	4473	214	108	164	130	85	35
U_V7.0_losses_34b45937	V7.0_losses	34b45937b42b7114	4473	214	108	164	130	85	16
U_V7.1_lsr_34b45937	V7.1_lsr	34b45937b42b7114	4473	214	108	164	130	85	6
U_V7.2R1_full_2f5a5c2e	V7.2R1_full	2f5a5c2e4832298a	4233	174	91	142	116	75	12
U_V7.2R2_trackB_e81801ff	V7.2R2_trackB	e81801ff29e01b7b	4219	174	91	142	115	75	12
U_V7.2R3_ppi_17487417	V7.2R3_ppi	174874178847c9bc	3452	132	70	105	87	55	15
U_V7.2R3_noise_17487417	V7.2R3_noise	174874178847c9bc	3452	132	70	105	87	55	12
U_V7.3_physics_f8c33c7e	V7.3_physics	f8c33c7e987d72c8	3351	166	86	129	102	65	30
U_V7.2_nested_c08fe062	V7.2_nested	c08fe062e64e4cba	1807	13	9	13	12	11	12

Supplementary Note 4 | Derivation of fixed-K shortlist discriminability

To separate candidate-set geometry from the realised magnitudes of $\Delta\Delta G$, we evaluate fixed-K selection against a *canonical rank-linear utility reference*. Rank the n candidates of a target from most to least favourable and assign the j-th ranked candidate the reference utility

$$u_{(j)} = n - j + 1, \qquad j = 1, \ldots, n, \tag{3}$$

so that utilities are equally spaced and strictly decreasing in rank. This reference is a deliberate idealisation: it fixes the *shape* of the utility profile and asks how much of the total utility a size-K shortlist can capture as a function of n and K alone. It is not a model of any particular $\Delta\Delta G$ distribution, and the resulting quantity is a geometric surrogate for shortlist discriminability rather than a property of real affinity data. Let $E = \sum_{i \in S_K} u_i / \sum_{i=1}^{n} u_i$ denote the share of total utility captured by a selected set S_K of K candidates. Two references define the discriminable interval:

- E_{max} — the share attained by the optimal (oracle) selection. Under Eq. (3) the top K candidates have utilities $n, n-1, \ldots, n-K+1$, and $\sum_{j=1}^{n} u_{(j)} = n(n+1)/2$, so

$$E_{\text{max}} = \frac{\sum_{j=1}^{K}(n-j+1)}{n(n+1)/2} = \frac{K\,(2n-K+1)}{n\,(n+1)}. \tag{4}$$

  The closed form depends on the reference Eq. (3); for an arbitrary utility vector $E_{\text{max}} = \sum_{j \le K} u_{(j)} / \sum_{j \le n} u_{(j)}$ and is not determined by n and K alone.

- E_{rand} — the share expected under uniform random selection of K of the n candidates. By symmetry of the uniform distribution over subsets, each candidate is selected with probability K/n irrespective of its utility, so

$$E_{\text{rand}} = \frac{K}{n}. \tag{5}$$

  This reference is exact under any utility profile, including the rank-linear one.

The discriminable interval is their difference, which simplifies in closed form:

$$D(n,K) \;=\; E_{\text{max}} - E_{\text{rand}} \;=\; \frac{K(2n-K+1)}{n(n+1)} - \frac{K}{n} \;=\; \frac{K\,(n-K)}{n\,(n+1)}. \tag{6}$$

For $n \le K$ every candidate is selected, hence $E_{\text{max}} = E_{\text{rand}} = 1$ and $D = 0$ identically: such targets carry no information for fixed-K shortlist metrics. Treating n as continuous and differentiating Eq. (6),

$$\frac{\mathrm{d}D}{\mathrm{d}n} = K\,\frac{(n+1) - (n-K)(2n+1)}{n^2(n+1)^2} = 0 \implies n - K = \frac{n(n+1)}{2n+1}, \tag{7}$$

whose positive root is

$$n^* = K + \sqrt{K^2 + K}. \tag{8}$$

For $K = 10$ this gives $n^* \approx 20.5$, with $D(20) = 0.2381$ and $D(50) = 0.1569$; in the present benchmark 53 of 130 targets with $n \ge 5$ (40.8%) fall in the degenerate regime $n \le K$.

Applicability and limits. Everything above is conditional on the canonical rank-linear reference Eq. (3). $D(n,K)$ is therefore a geometric surrogate for shortlist discriminability, not a statement about arbitrary utility distributions: with any other utility profile E_{max} changes and so does D. It is not a model-performance metric, it does not measure the $\Delta\Delta G$ utility range of any real dataset, and it does not imply that SKEMPI $\Delta\Delta G$ values are intrinsically similar. Establishing that would require an analysis using realised utility magnitudes, which we did not perform. What is reference-free is the degenerate case: for $n \le K$ a size-K shortlist selects every candidate, so $E_{\text{max}} = E_{\text{rand}} = 1$ and $D = 0$ under *any* utility profile. That statement, and the target counts it implies, does not depend on Eq. (3).

Supplementary Note 5 | Weighted covariance decomposition

The five target-weighting schemes act on the *target level*: predictions are unchanged and only the weight assigned to each target varies. Let n_c be the number of mutations in target c. Because the schemes are named after target-aggregation rules, we take the target weight A_c as the quantity they describe,

$$A_c \in \{\, n_c,\ 1,\ \sqrt{n_c},\ \log(1+n_c),\ \min(n_c, 30)\,\} \tag{9}$$

for the per-mutation, per-target, $\sqrt{n}$, logarithmic and capped schemes respectively, and split it evenly across the target's rows:

$$w_{ci} = \frac{A_c}{n_c}, \qquad \sum_i w_{ci} = A_c, \qquad A = \sum_c \sum_i w_{ci} = \sum_c A_c. \tag{10}$$

This is the only reading under which the scheme names are accurate: with $A_c = 1$ each target carries total weight 1 regardless of its size (genuinely per-target), and with $A_c = n_c$ every row carries weight 1 (genuinely per-mutation). Writing $w_{ci} = a_c$ instead — the literal form present in the original implementation — would give each target total weight $n_c a_c$, so that the "per-target" scheme would still let large targets dominate; we therefore treat that literal reading as a sensitivity variant (Eq. (17) below).

Under the measure in Eq. (10) the weighted means and variances are

$$\bar{y}_w = \frac{1}{A} \sum_c \sum_i w_{ci}\, y_{ci}, \qquad \sigma^2_{y,w} = \frac{1}{A} \sum_c \sum_i w_{ci}\, (y_{ci} - \bar{y}_w)^2, \tag{11}$$

and likewise for $\hat{y}$. Because $w_{ci} = A_c/n_c$ is constant within a target, the within-target mean is the ordinary target mean,

$$B_{y,c} \;=\; \frac{\sum_i w_{ci}\, y_{ci}}{\sum_i w_{ci}} \;=\; \bar{y}_c. \tag{12}$$

With $\varepsilon_{ci} = y_{ci} - B_{y,c}$ and $\hat{\varepsilon}_{ci} = \hat{y}_{ci} - B_{\hat{y},c}$, the components are

$$C_{\mathrm{B},w} = \frac{\sum_c A_c \left(B_{y,c} - \bar{y}_w\right)\left(B_{\hat{y},c} - \bar{\hat{y}}_w\right)}{A\, \sigma_{y,w}\, \sigma_{\hat{y},w}}, \tag{13}$$

$$C_{\mathrm{W},w} = \frac{\sum_c \sum_i w_{ci}\, \varepsilon_{ci}\, \hat{\varepsilon}_{ci}}{A\, \sigma_{y,w}\, \sigma_{\hat{y},w}}. \tag{14}$$

Substituting $y_{ci} = B_{y,c} + \varepsilon_{ci}$ and $\hat{y}_{ci} = B_{\hat{y},c} + \hat{\varepsilon}_{ci}$ into $A\, \sigma_{y,w}\sigma_{\hat{y},w}\, r_w = \mathrm{Cov}_w(y, \hat{y})$ gives

$$\sum_c \sum_i w_{ci}\, (y_{ci} - \bar{y}_w)(\hat{y}_{ci} - \bar{\hat{y}}_w) = \underbrace{\sum_c A_c \big(B_{y,c} - \bar{y}_w\big)\big(B_{\hat{y},c} - \bar{\hat{y}}_w\big)}_{\text{Eq. (13)}} + \underbrace{\sum_c \sum_i w_{ci}\, \varepsilon_{ci} \hat{\varepsilon}_{ci}}_{\text{Eq. (14)}}, \tag{15}$$

because for each target $\sum_i w_{ci}\varepsilon_{ci} = 0$ and $\sum_i w_{ci}\hat{\varepsilon}_{ci} = 0$, so both cross terms vanish. Hence

$$r_w \;=\; C_{\mathrm{B},w} + C_{\mathrm{W},w} \tag{16}$$

holds exactly for every scheme and every evaluation instance. Verified numerically on all 109 configurations crossed with all five schemes (545 instances): maximum absolute residual 5.6×10^{-17}.

Sensitivity to the weighting reading. If instead the row weight is taken literally as $w_{ci} = a_c$ with a_c given by Eq. (9), the target total weight becomes $n_c a_c$ and the same algebra applies with

$$w^{\mathrm{sens}}_{ci} = a_c, \qquad \sum_i w^{\mathrm{sens}}_{ci} = n_c\, a_c. \tag{17}$$

The identity still holds exactly, because numerator and denominator share one measure. The two readings differ only in *what a scheme equalises*, and therefore in the size of the reported components:

Quantity	Eq. (10) (primary)	Eq. (17) (sensitivity)
Median cross-scheme range of C_{B}	0.3052	0.0930
Median cross-scheme range of C_{W}	0.1240	0.0624
C_{B} sign reversals (any magnitude)	40/109	44/109
C_{B} sign reversals with $\lvert C_{\mathrm{B}}\rvert > 0.005$	34/109	32/109
C_{W} sign reversals (any magnitude)	5/109	0/109

Both readings agree on the qualitative conclusion: the between-target component is markedly less stable across weighting conventions than the within-target component (a larger cross-scheme range and many more sign reversals), so a reported decomposition must state its weighting scheme. They disagree on whether C_{W} ever reverses sign (5/109 under the primary reading versus 0/109 under the sensitivity reading), and on the absolute magnitude of the ranges; the main text quotes the primary reading and reports the sensitivity here.

Supplementary Note 6 | Duplicate-key sensitivity

The controlled-split universe contains 7,523 mutation records but only 4,812 distinct (complex, mutation) keys: 2,011 keys carry more than one experimental record, and 1,405 of those disagree in $\Delta\Delta G$. This raises a legitimate concern: could part of the random-split performance, and hence part of the apparent collapse under family-out-of-domain evaluation, be an artefact of the same key appearing more than once in the evaluation set?

Because the stored prediction artifacts do not retain per-row fold assignments, the split cannot be regenerated here; we therefore quantify the sensitivity *within* the stored evaluation set by recomputing every quantity under three key treatments:

- **ALL** — all 7,523 records (the reported analysis);
- **FIRST** — one record per key, keeping the first occurrence (4,812 records);
- **MEAN** — one record per key, with y and $\hat{y}$ both averaged within the key (4,812 records).

The decomposition used here is the unweighted identity of Supplementary Note 1, and was verified to reproduce the frozen per-configuration values to 3.3e-16 across the 36 controlled-split configurations.

Regime	Key treatment	Median r	Median C_{B}	Median C_{W}	Median macro ρ
Random	ALL	+0.6844	+0.4275	+0.2591	0.0707
Random	FIRST	+0.6378	+0.3920	+0.2465	0.0886
Random	MEAN	+0.6338	+0.3705	+0.2647	0.0977
Family-OOD	ALL	+0.1508	−0.0072	+0.1580	0.0377
Family-OOD	FIRST	+0.1614	−0.0122	+0.1759	0.0354
Family-OOD	MEAN	+0.2181	+0.0197	+0.1956	0.0347

The paired Random → Family-OOD differences are:

Key treatment	Δr	ΔC_{B}	ΔC_{W}	Between-target share
ALL	-0.5332	-0.4324	-0.1011	81.0 %
FIRST	-0.4733	-0.4027	-0.0706	85.1 %
MEAN	-0.4266	-0.3476	-0.0671	83.8 %

Three observations follow. First, collapsing duplicate keys *reduces* the random-split correlation (+0.6844 to +0.6378 under FIRST) while slightly *increasing* the family-out-of-domain correlation (+0.1508 to +0.1614); the collapse is therefore, if anything, understated rather than inflated by repeated records. Second, ΔC_{B} remains large and negative under every treatment, and its share of the summed absolute component change lies between 81.0 % and 85.1 %: the qualitative conclusion that the between-target component carries most of the decline does not depend on how repeated keys are handled. Third, the magnitude of the decline is sensitive to the

treatment (-0.5332 versus -0.4266), so the absolute size of the collapse should be quoted with the key treatment stated.

Limits. This is an evaluation-set-internal sensitivity analysis. It cannot rule out duplicate-key leakage across the train/test boundary, because that requires regenerating the splits under key-level grouping and retraining, which the stored artifacts do not permit and which we did not perform. The 1,405 keys with disagreeing $\Delta\Delta G$ values also indicate that a (complex, mutation) key does not always identify a unique experimental condition; determining whether those conditions should enter the key definition is a data-curation decision that we did not make.

Supplementary Note 7 | External coverage and common intersection

The five external methods are annotation columns of the same published SKEMPI table, mapped to the annotation namespace through the normalised key of `scripts/v8_1/05_external_clean.py`. Each method has its own pattern of missing entries, so the uniquely mapped subsets differ slightly in size: Flex-ddG maps 1,755 rows, DDGPred 1,756, and FoldX, ESM-IF and proteinMPNN 1,776, out of 1,781 uniquely mappable rows. Because all five columns originate from the same annotation universe and share common row identifiers, we can remove the coverage imbalance by restricting every method to their common row intersection.

Method	Rows	Targets ($n \geq 5$)	Pooled Pearson	Macro Spearman	C_{B}	C_{W}	RMSE
(A) variable coverage — each method's own uniquely mapped subset, raw columns							
DDGPred	1,756	65	+0.2406	+0.1287	+0.0715	+0.1690	2.1262
ddG_FoldX	1,776	66	+0.2089	+0.1088	+0.0657	+0.1433	2.3691
Flex-ddG	1,755	65	+0.2028	+0.1092	+0.0478	+0.1550	2.1636
proteinMPNN	1,776	66	+0.1509	+0.0320	+0.0699	+0.0810	—
ESM-IF	1,776	66	-0.2250	+0.0213	-0.1149	-0.1101	—
(A′) same subsets, exploratory sign-oriented values							
DDGPred	1,756	65	+0.2406	+0.1287	+0.0715	+0.1690	2.1262
ddG_FoldX	1,776	66	+0.2089	+0.1088	+0.0657	+0.1433	2.3691
Flex-ddG	1,755	65	+0.2028	+0.1092	+0.0478	+0.1550	2.1636
proteinMPNN	1,776	66	+0.1509	+0.0320	+0.0699	+0.0810	—
ESM-IF	1,776	66	+0.2250	-0.0213	+0.1149	+0.1101	—
(B) common intersection — raw columns							
DDGPred	1,752	65	+0.2410	+0.1287	+0.0714	+0.1696	2.1265
ddG_FoldX	1,752	65	+0.2121	+0.0967	+0.0681	+0.1440	2.3623
Flex-ddG	1,752	65	+0.2036	+0.1092	+0.0476	+0.1560	2.1620
proteinMPNN	1,752	65	+0.1497	+0.0430	+0.0690	+0.0807	—
ESM-IF	1,752	65	-0.2225	+0.0099	-0.1113	-0.1112	—
(B′) common intersection, exploratory sign-oriented values							
DDGPred	1,752	65	+0.2410	+0.1287	+0.0714	+0.1696	2.1265
ddG_FoldX	1,752	65	+0.2121	+0.0967	+0.0681	+0.1440	2.3623
Flex-ddG	1,752	65	+0.2036	+0.1092	+0.0476	+0.1560	2.1620
proteinMPNN	1,752	65	+0.1497	+0.0430	+0.0690	+0.0807	—
ESM-IF	1,752	65	+0.2225	-0.0099	+0.1113	+0.1112	—

RMSE is shown only for Flex-ddG, FoldX and DDGPred, whose columns are on a $\Delta\Delta G$-like scale; ESM-IF and proteinMPNN return dimensionless mutation-effect scores for which an error in kcal/mol is not defined (Supplementary Note 8).

The common intersection retains 1,752 of the 1,781 uniquely mappable rows (98.4 %), covering 163 targets of which 65 have $n \geq 5$. Comparing the treatments:

- on the *raw* columns the two rankings are strongly positively aligned (Spearman $\rho = +0.90$, exact permutation $p = 0.083$), as in the main text;
- after the exploratory sign-orientation the same ranks show no significant agreement ($\rho = +0.30$, $p = 0.683$), and ESM-IF moves from last to second on the pooled axis while remaining last on the within-target axis;
- in both treatments DDGPred ranks first on both axes.

The coverage imbalance is therefore not what drives the difference between the two conclusions: restricting all five methods to their common intersection leaves the raw panel aligned and the oriented panel unaligned, exactly as in the full uniquely mapped subsets. The panel remains small (five methods) and underpowered under either treatment.

Supplementary Note 8 | Provenance, orientation and units of the external predictions

The five external methods are not separate downloads: they are five annotation columns of the SKEMPI-derived protein–protein interaction table distributed with CATH-ddG (main text reference 3; `skempi_v2.csv`), which we read directly without modification (Table S3). This matters for interpretation in three ways.

The published columns are preserved unmodified. For every method we ingest the published column exactly as distributed. We apply no affine rescaling and no recalibration, and the raw values reproduce to four decimals when recomputed directly from the published columns on the same uniquely mapped subset (e.g. ESM-IF -0.2250 and ProteinMPNN $+0.1509$). The only operation applied anywhere is the single sign multiplier o_m introduced in the main text Methods (*Score orientation and its limits*). It is used solely for the exploratory orientation sensitivity and never modifies the raw columns reported here; because o_m is selected from the evaluation data, that sensitivity is explicitly label-informed, whereas the raw analysis requires no such choice and is the primary result. So the reported raw correlations are properties of the published scores and not of any post-processing.

Orientation. The analysed response is $y = \Delta\Delta G = \Delta G_{\text{mut}} - \Delta G_{\text{WT}}$ (Methods), so $y > 0$ denotes weaker binding. We use every column exactly as published and apply no sign inversion to any method. Two consequences are worth stating. First, the Flex-ddG, FoldX and DDGPred columns are already oriented so that larger values mean weaker binding, matching y, and they correlate positively with it. Second, the published ESM-IF column is oriented the other way — larger values mean stronger binding — which is why it carries a negative pooled correlation (-0.2250). That negative sign is a property of the published ESM-IF score on this benchmark, not an artefact of our processing; we did not flip it, and we flag it here so that the sign is not misread.

Units, and why two RMSE values are not reported. Flex-ddG, FoldX and DDGPred are quantitative $\Delta\Delta G$-like predictors (signed binding-affinity changes) whose columns span standard deviations of 1.31, 1.92 and 1.55 respectively, so an error in kcal/mol is meaningful for them. ESM-IF and ProteinMPNN are inverse-folding sequence models: their published outputs are dimensionless mutation-effect scores with standard deviations of 0.093 and 0.012, roughly two orders of magnitude narrower than the experimental $\Delta\Delta G$ spread. An RMSE expressed in kcal/mol is therefore not defined for those two columns, and we report "—". This follows the rule already applied elsewhere in this work, where a dimensionless mutation-effect score is recorded with `RMSE = NOT_APPLICABLE` rather than converted into an apparent energy error. Correlation-based quantities — pooled Pearson, macro within-target Spearman, C_{B} and C_{W} — are invariant to positive affine rescaling and are reported for all five methods.

Sign orientation for cross-method comparison. The published columns are not all in the same direction. Using within-complex correlations against $y = \Delta\Delta G$ — which removes between-target differences and therefore isolates direction rather than predictive skill — Flex-ddG, FoldX, DDGPred and ProteinMPNN are aligned with y ($+0.49$, $+0.49$, $+0.51$ and $+0.30$), whereas ESM-IF is anti-aligned (-0.24): its published score is larger for stronger binding. For the cross-method performance comparison we therefore apply a sign-only orientation,

$s^{\text{aligned}} = o_m\, s^{\text{raw}}$ with $o_{\text{ESM-IF}} = -1$ and $o_m = +1$ for the other four (Methods). This uses no evaluation labels and no recalibration, and it changes no information content: it only flips the sign of the correlation-like quantities. Table S3 records each raw column with its raw orientation and o_m, so the unmodified provenance and the aligned values are both auditable. The effect on the panel is material: used raw the two axes appear strongly aligned ($\rho = +0.90$, $p = 0.083$), whereas after orientation they show no significant agreement ($\rho = +0.30$, $p = 0.683$). We report the oriented values throughout and state the raw alternative explicitly, since with five methods neither reaches adequate power.

Scope of the comparison. Because the two score-based methods are not on the same scale as the three energy-based ones, the RMSE column in Table 3 should not be used to compare across those groups; the pooled and within-target correlations are the quantities that are comparable across all five methods, and they are the basis of the rank-agreement analysis.

Machine-readable supplementary data

The full model-family and configuration registry — including per-configuration row-set fingerprints, split regimes, training-provenance flags, ablation and representative markers, and the representative-selection reason — is supplied as machine-readable Supplementary Data (`MODEL_FAMILY_REGISTRY.csv`, `MATCHED_UNIVERSE_REGISTRY.csv`, `FROZEN_INPUT_MANIFEST.csv`). The weighting audit configurations are enumerated in `WEIGHTING_FINAL.csv`, and the duplicate-key sensitivity in `DUPLICATE_KEY_SENSITIVITY.csv`.

Supplementary Figures

Table S2: Model pools used in the rank-agreement sensitivity analysis.

Row-set hash	Level	Models	Method families	Green models	ρ(pooled, macro)
f8c33c7e	A	30	10	30	+0.8242
8d003e00	A	42	12	12	+0.7596
8d003e00	C	12	12	3	+0.6923
c08fe062	A	12	4	12	+0.5315
2f5a5c2e	A	12	4	12	+0.4336
e81801ff	A	12	4	12	+0.3427
17487417	C	9	9	9	+0.0769
17487417	A	27	9	27	-0.0174
34b45937	C	34	34	27	-0.0716
34b45937	A	67	34	60	-0.2685

Level A (∗LA in Figure S3) denotes a provenance-clean stratum: every listed model is free of known supervised training overlap with the evaluation rows. *Level C* (∗LC) denotes a permissive-split stratum, where the pool includes configurations trained under random/complex/`seq40` partitions whose evaluation rows may overlap their training folds; these are retained only for this sensitivity analysis. 'Green models' is the number of models in the pool flagged provenance-clean. Pools within the same row-set hash are comparable because they share identical evaluation rows; pools with different hashes are not.

Table S3: Provenance and score semantics of the external prediction columns.

Method	Raw column	Units	Raw orientation	o_m	RMSE meaningful?
Flex-ddG	`Flex-ddG`	kcal/mol (ΔΔG-like scaling)	higher = larger $\Delta\Delta G$ (weaker binding)	$+1$	yes
DDGPred	`DDGPred`	kcal/mol (ΔΔG-like scaling)	higher = larger $\Delta\Delta G$ (weaker binding)	$+1$	yes
ddG_FoldX	`ddG_FoldX`	kcal/mol (ΔΔG-like scaling)	higher = larger $\Delta\Delta G$ (weaker binding)	$+1$	yes
ESM-IF	`ESM-IF`	dimensionless (no kcal/mol scaling)	higher = stronger binding	-1	**no**
proteinMPNN	`proteinMPNN`	dimensionless (no kcal/mol scaling)	higher = larger $\Delta\Delta G$ (weaker binding)	$+1$	**no**

All five columns come from one published annotation table of SKEMPI-derived protein–protein interaction mutations. Raw values are used exactly as published; we apply no affine calibration and no recalibration against the evaluation labels. The o_m column records the sign multiplier used only for the exploratory orientation sensitivity. We verified this by recomputing every reported correlation directly from the raw columns, which reproduces the reported values to four decimals for all five methods. Raw standard deviations differ by more than an order of magnitude across columns (1.31, 1.55 and 1.92 for the three $\Delta\Delta G$-like predictors versus 0.093 and 0.012 for the two inverse-folding scores), which is why an RMSE in kcal/mol is defined only for the former.

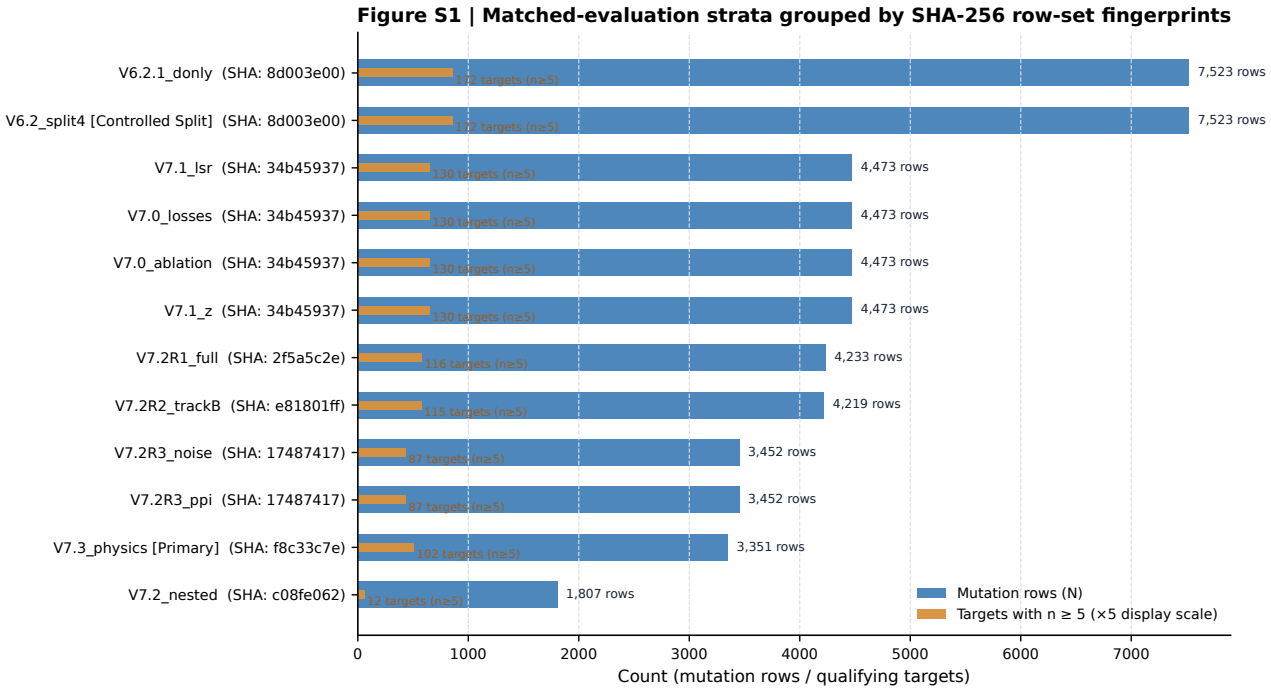


Figure S1: Matched-evaluation strata grouped by SHA-256 row-set fingerprints. Each bar is one prediction output; outputs sharing a fingerprint are directly comparable with each other and with no others.

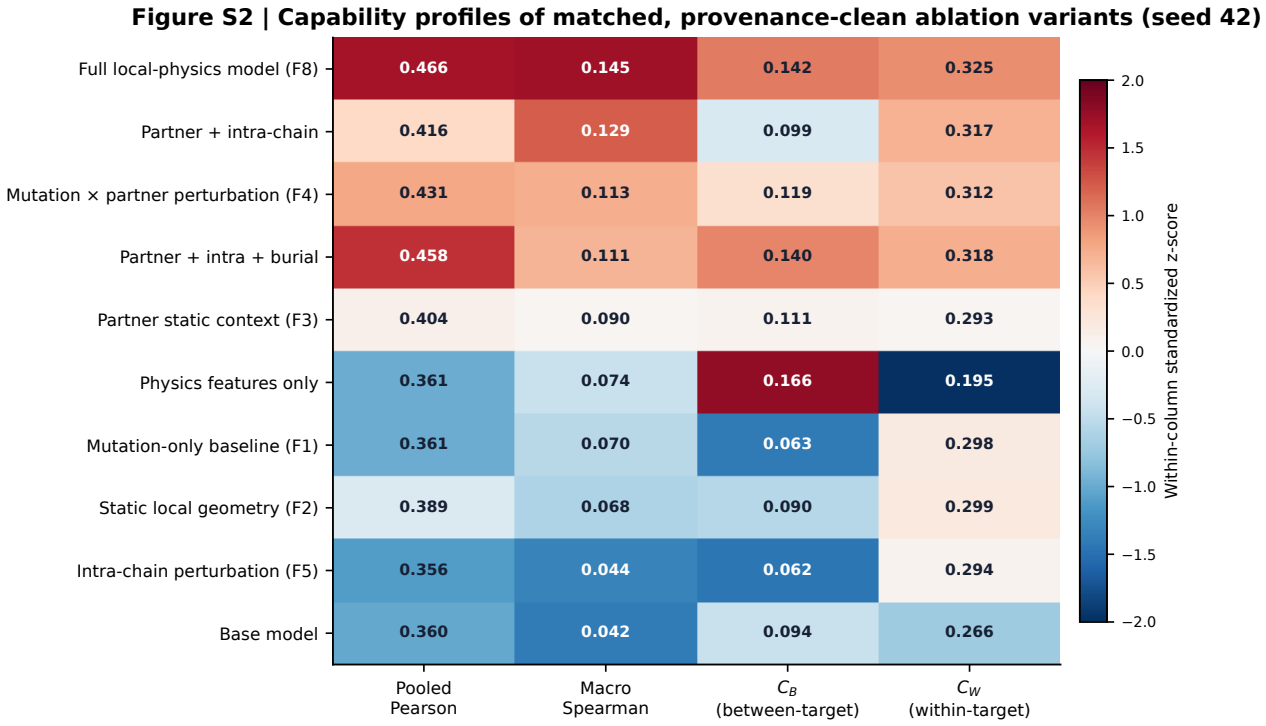


Figure S2: Capability profiles of matched, provenance-clean ablation variants (seed 42). Rows are controlled ablation variants of one local-physics method family, not independent methods. Values are standardised within each metric for visual comparison only; raw values are printed in each cell.

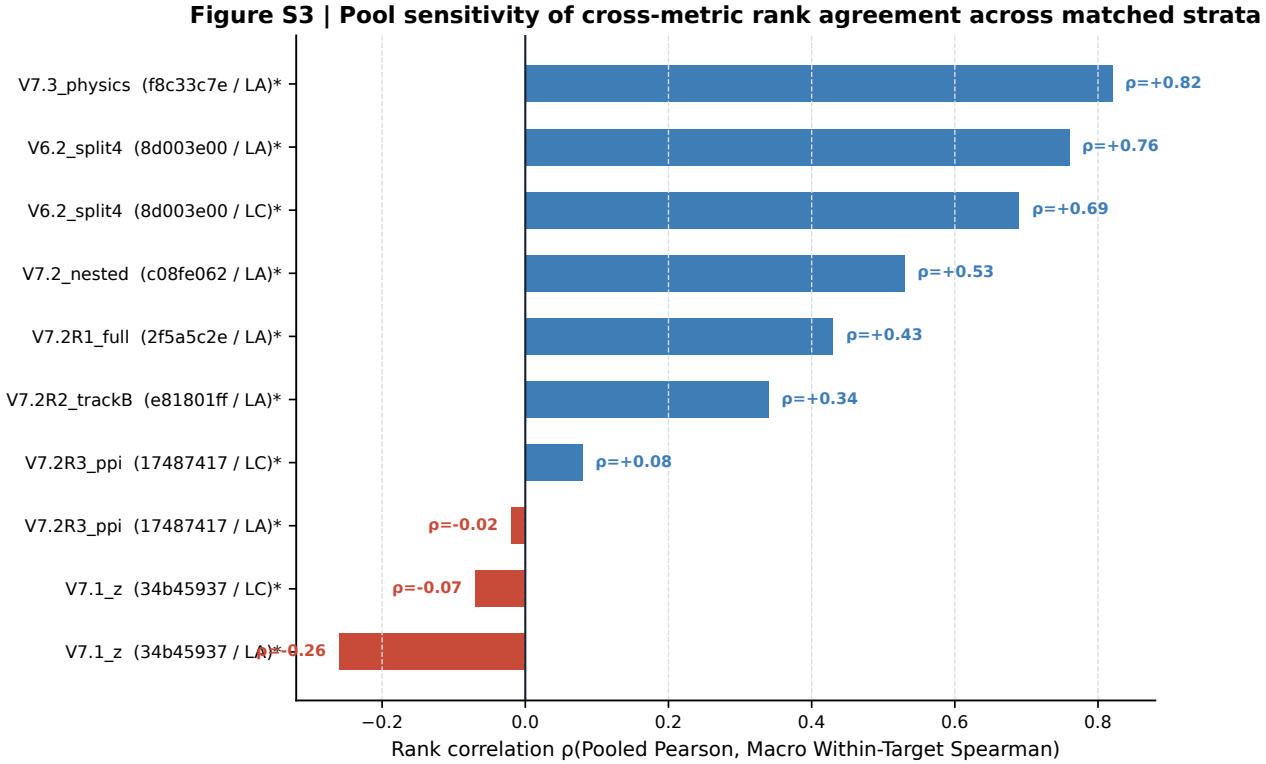


Figure S3: Pool sensitivity of cross-metric rank agreement across matched strata. Each bar is one model pool within one row-set stratum, labelled `artifact (row-set hash / level)`. *LA* denotes a provenance-clean stratum (no known supervised training overlap) and *LC* a permissive-split stratum retained only for this sensitivity analysis; pool sizes, method-family counts and inclusion criteria are listed in Table S2 below. Within one row-set hash the evaluation rows are identical and only the set of included model families changes, which moves the agreement between pooled Pearson and macro within-target Spearman from -0.26 to $+0.82$; this quantity is therefore reported as a range rather than a single value.